3.1

2. WoLog $a=0$ Let $|z| = s < r \Rightarrow \sum_0^\infty |a_n||z|^n, \sum_0^\infty |b_n||z|^n$ conv.

$\therefore$ for any $N \in \mathbb{Z}^+$ $\sum_{n=0}^N |a_n + b_n| |z|^n \leq \sum_{n=0}^N |a_n| |z|^n + \sum_{n=0}^N |b_n||z|^n$

$\Rightarrow \sum_{n=0}^\infty (a_n+b_n) z^n$ is abso. conv. Since z was arbitrary in

$B(0,r) \Rightarrow$ rad. conv. $\sum_{n=0}^\infty (a_n+b_n) z^n \geq r$

also for any $N \in \mathbb{Z}^+$

$$\sum_{n=0}^N (a_n+b_n) z^n = \sum_{n=0}^N a_n z^n + \sum_{n=0}^N b_n z^n$$

hence

$$\sum_{n=0}^\infty (a_n+b_n) z^n = \sum_{n=0}^\infty a_n z^n + \sum_{n=0}^\infty b_n z^n$$

3. $\{a_n\}, \{b_n\}$ bounded. Let $n \in \mathbb{Z}^+$ and let $m \geq n$

$$a_m + b_m \leq \sup \{a_n, a_{n+1}, a_{n+2}, \dots\} + \sup\{b_n, b_{n+1}, b_{n+2}, \dots\}$$

$$\Rightarrow \sup\{a_n+b_n, a_{n+1}+b_{n+1}, a_{n+2}+b_{n+2}, \dots\} \leq \sup\{a_n, a_{n+1}, a_{n+2}, \dots\} + \sup\{b_n, b_{n+1}, b_{n+2}, \dots\}$$

$$\Rightarrow \limsup\{a_n+b_n\} \leq \sup\{a_n, a_{n+1}, a_{n+2}, \dots\} + \sup\{b_n, b_{n+1}, b_{n+2}, \dots\}$$

$$\Rightarrow \limsup\{a_n+b_n\} \leq \limsup\{a_n\} + \limsup\{b_n\}$$

4. Let $n \in \mathbb{Z}^+$, $m \geq n$ Then $a_m \leq \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\}$

$$\Rightarrow \inf\{a_m, a_{m+1}, a_{m+2}, \cdots\} \leq \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\}$$

$$\Rightarrow \liminf\{a_n\} \leq \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\}$$

$$\Rightarrow \liminf\{a_n\} \leq \limsup\{a_n\}$$

5. Since $\lim a_n = a$ $\forall \varepsilon > 0$ $\exists N \ni n > N$

$$a - \varepsilon < a_n < a + \varepsilon$$

$$\Rightarrow a - \varepsilon < \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\} \leq a + \varepsilon$$

$$\Rightarrow a - \varepsilon \leq \limsup\{a_n\} \leq a + \varepsilon$$ Since ε arbitrary

$$\Rightarrow a = \limsup\{a_n\}$$

6 a) $\sum c^n z^n$ $\quad c=0 \Rightarrow R=\infty$

$c\neq 0 \Rightarrow |a_n|^{\frac{1}{n}} = |c| \Rightarrow R=\frac{1}{|c|}$

b) $\sum a^{n^2} z^n$ $\quad |a_n|^{\frac{1}{n}} = |a|^n$ $\quad |a|>1 \Rightarrow R=0$

$|a|=1 \Rightarrow R=1$

$|a|<1 \Rightarrow R=\infty$

c) Subcase of a) $R=\frac{1}{|k|}$

d) $\sum_{n=0}^{\infty} z^{n!}$ $\quad$ inf. many of the $a_n = 1 \Rightarrow$

infinitely many $|a_n|^{\frac{1}{n}} = 1 \Rightarrow$

$\limsup |a_n|^{\frac{1}{n}} = 1 \Rightarrow R=1$

7. $\left[\frac{1}{n(n+1)}\right]^{\frac{1}{n(n+1)}} \leq \left[\frac{1}{n}\right]^{\frac{1}{n(n+1)}} \leq 1$ $\quad$ By Sandwich Theorem & Prob 3 of 3.2

$\limsup |a_n|^{\frac{1}{n}} = 1$

at $z=1$ $\sum \frac{(-1)^n}{n}$ converges conditionally

at $z=-1$ $\sum \frac{(-1)^n}{n}$ converges conditionally

at $z=i$ $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} (i)^{n(n+1)} = \frac{1}{1} - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \cdots$

converges conditionally by Dirichlet test

3.2

1. $f(z) = |z|^2$ diff at 0 only

at $z=0$ $\lim\limits_{h\to 0} \frac{f(z+h)-f(z)}{h} = \lim\limits_{h\to 0} \frac{|h|^2}{h} = \lim\limits_{h\to 0} \bar{h} = 0$

at $z\neq 0$ $\lim\limits_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{f(z+h)-f(z)}{h} = \lim\limits_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{|z+h|^2-|z|^2}{h} = \lim\limits_{\substack{h\to 0\\ h\in\mathbb{R}}} \frac{(x+h)^2+\cancel{y^2}-x^2-\cancel{y^2}}{h} = 2x$

$z = x+iy$ $\neq$

$\lim\limits_{\substack{h\to 0\\ h=ig\\ g\in\mathbb{R}}} \frac{f(z+h)-f(z)}{h} = \lim\limits_{\substack{g\to 0\\ g\in\mathbb{R}}} \frac{|z+ig|^2-|z|^2}{ig} = \lim\limits_{\substack{y\to 0\\ y\in\mathbb{R}}} \frac{\cancel{x^2}+(g+y)^2-x^2-y^2}{ig} = \frac{2y}{i}$

3. $\lim n^{\frac{1}{n}} = y \iff \lim \log n^{\frac{1}{n}} = \log y$

$\ln y = \lim \frac{\log n}{n} = 0 \Rightarrow y = 1$

4. $\cos z = \sum\limits_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$ $(\cos z)' = \sum\limits_{n=1}^{\infty} \frac{(-1)^n 2n z^{2n-1}}{(2n)!} = \sum\limits_{n=1}^{\infty} (-1)^n \frac{z^{2n-1}}{(2n-1)!}$

$R = \infty$

$1+m = n$

$= \sum\limits_{m=0}^{\infty} (-1)^{m+1} \frac{z^{2m+1}}{(2m+1)!}$

$\sin z = \sum\limits_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$

$= -\sin z$

6. $\{z \mid e^z = i\}$ $\quad |i| = 1 \Rightarrow |e^z| = 1 \Rightarrow \mathrm{Re}\, z = 0$

$i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \Rightarrow \arg i = \frac{\pi}{2} + 2\pi k$ for $k \in \mathbb{Z}$

$\Rightarrow \mathrm{Im}\, z = \arg i$

$\Rightarrow \{z \mid e^z = i\} = \{z = 0 + i(\frac{\pi}{2} + 2\pi k) \mid k \in \mathbb{Z}\}$

$\{z \mid e^z = -1\}$ similar, $\{z \mid e^z = -i\}$ similar

$\{z \mid \cos z = 0\}$ $\quad \cos z = \cos x + iy = \cos x \cosh y - i \sin x \sinh y$

hence $\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi k$ for $k \in \mathbb{Z}$

$\sinh y = 0 \Rightarrow y = 0$

$\{z \mid \cos z = 0\} = \{\frac{\pi}{2} + \pi k + 0i \mid k \in \mathbb{Z}\}$

$\{z \mid \sin z = 0\}$ similar

fix a and define

7. $g(z) = \cos(z)\cos(a-z) - \sin z \sin(a-z)$

$g'(z) = \cos z\left(-\cancel{\sin(a-z)(-1)}\right) - \sin z \cancel{\cos(a-z)} - \cos z \cancel{\sin(a-z)} - \sin z \cancel{\cos(a-z)(-1)}$

$= 0 \Rightarrow g(z) \equiv c = g(0) = \cos a$

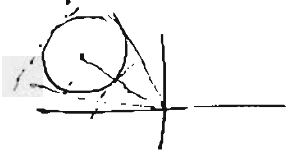
let $a = z+w \Rightarrow \cos(z+w) = \cos z \cos w - \sin z \sin w$

8. $\tan z \in \mathcal{H}(\mathbb{C} \setminus Z_{\cos z}) = \mathcal{H}(\mathbb{C} \setminus \{\frac{\pi}{2} + \pi k \mid k \in \mathbb{Z}\})$

9. $z_n, z \in \mathbb{C} \setminus (-\infty, 0]$ $\quad z_n = r_n e^{i\theta_n},\ z = r e^{i\theta}$ $\quad -\pi < \theta_n, \theta < \pi$

$z_n \to z \Rightarrow \exists\, \delta \ni B(z,\delta) \cap (-\infty, 0] = \emptyset$

$\Rightarrow \exists\, N \ni n > N \quad z_n \in B(z,\delta)$



$\Rightarrow n > N \quad |r_n - r| < \delta$

$\Rightarrow n > N \quad |\theta_n - \theta| < \arcsin\frac{\delta}{r} < \frac{\pi}{2}\frac{\delta}{r}$

10. SKIP

11. $n=1$ $\quad z^1 = z$ and $e^{1 f(z)} = e^{f(z)} = z$

assume $n=k$ $\quad z^k = e^{k f(z)}$

$n=k+1$ $\quad z^{k+1} = z^k z$ and $e^{(k+1) f(z)} = e^{k f(z) + f(z)} = e^{k f(z)} e^{f(z)} = z^k z$

$n=0$ $\quad z^0 \equiv 1$ and $e^{0 f(z)} = e^0 = 1$

$n<0$, $n=-m$ $\quad z^n = z^{-m} = \frac{1}{z^m}$ and $e^{n f(z)} = e^{-m f(z)} = \frac{1}{e^{m f(z)}} = \frac{1}{z^m}$

12. $\operatorname{Re} z^{\frac{1}{2}} = \operatorname{Re} e^{\frac{1}{2}\log z}$ where $z \in \mathbb{C} \setminus (-\infty, 0] \Rightarrow$

$-\pi < \arg z < \pi$

$-\frac{\pi}{2} < \frac{1}{2}\arg z < \frac{\pi}{2}$

$= \operatorname{Re} e^{\frac{1}{2}\log|z| + i\frac{1}{2}\arg z}$

$= \operatorname{Re} \sqrt{|z|}\, e^{i\frac{1}{2}\arg z}$

$= \sqrt{|z|} \cos\left(\frac{1}{2}\arg z\right)$ but $\cos \frac{1}{2}\arg z > 0$

14 $f \in H(G)$, G connected f is real $\Rightarrow \operatorname{Im} f = 0 \Rightarrow \operatorname{Im} f$ constant

$f(z) = u(z) + i v(z)$ where $v(z) = c \Rightarrow \begin{cases} v_x = 0 & v_x = -u_y \Rightarrow u_y = 0 \\ v_y = 0 & v_y = u_x \Rightarrow u_x = 0 \end{cases}$

$\Rightarrow f'(z) = u_x(z) + i v_x(z) \equiv 0$

$\Rightarrow f$ constant

15. $A = \{ w \mid w = \exp(\frac{1}{z}) \text{ s.t. } z \in B'(0,r) \}$

Let $B = \{ \zeta \mid \zeta = \frac{1}{z} \text{ s.t. } z \in B'(0,r) \} = \mathbb{C} \setminus \overline{B(0, \frac{1}{r})}$

$\Rightarrow S = \{ \zeta \mid \frac{1}{r} \le \operatorname{Im} \zeta \le \frac{1}{r} + 2\pi \} \subset B$

$\exp : S \to \mathbb{C} \setminus \{0\} \Rightarrow \exp : B \to \mathbb{C} \setminus \{0\}$

17. $f(z) = \sqrt{1-z} \Rightarrow 1-z \in \mathbb{C} \setminus (-\infty, 0]$

$\Rightarrow z \in \mathbb{C} \setminus [1, \infty) = G$

for $z \in G$ $\sqrt{1-z} = e^{\frac{1}{2} \log(1-z)} = e^{\frac{1}{2} \log|1-z| + i \frac{1}{2} \arg(1-z)}$

$= \sqrt{|1-z|}\, e^{i \frac{1}{2} \arg(1-z)}$

$-\pi < \arg(1-z) < \pi$

19. G region $\quad G^* = \{z \mid \bar{z} \in G\}$ $\quad f \in \mathcal{A}(G)$

$f^*(z) = \overline{f(\bar{z})}$ for $z \in G^*$

Let $z \in G^* \Rightarrow \exists\, \delta \ni B(\bar{z}, \delta) \subset G$

let $|h| < \delta$ $\quad \displaystyle\lim_{h \to 0} \frac{f^*(z+h) - f^*(z)}{h} = \lim_{h \to 0} \frac{\overline{f(\overline{z+h})} - \overline{f(\bar{z})}}{h}$

$$= \lim_{h \to 0} \overline{\left[\frac{f(\bar{z}+\bar{h}) - f(\bar{z})}{\bar{h}}\right]} = \overline{f'(\bar{z})}$$

21. Sp $\exists\, l$ on $\mathbb{C}\setminus\{0\}$ which is branch of $\log z$

Then l is continuous on $\mathbb{C}\setminus\{0\}$ $\Rightarrow \displaystyle\lim_{\substack{\theta \to \pi \\ \theta < \pi}} l(e^{i\theta}) = \lim_{\substack{\theta \to -\pi \\ \theta > -\pi}} l(e^{i\theta})$

But l is defined on $\mathbb{C}\setminus(-\infty, 0]$ and

satisfies conditions to be branch of $\log z$ on $\mathbb{C}\setminus(-\infty, 0]$

$\therefore\ l \equiv \log z + 2\pi k i$ for ~~some~~ fixed k

but $\displaystyle\lim_{\substack{\theta \to \pi \\ \theta < \pi}} \log e^{i\theta} + 2\pi k i = \pi i + 2\pi k i$

$\displaystyle\lim_{\substack{\theta \to -\pi \\ \theta > -\pi}} \log e^{i\theta} + 2\pi k i = -\pi i + 2\pi k i$ $\quad \rightarrow\!\leftarrow$